

L3 Gate 3 SSG-8 Mersenne RIGOROUS Closure

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L3 Gate 3 SSG-8 Mersenne RIGOROUS v0.1

0. Goal

Per Casey Thursday PM long task set L3: Gate 3 SSG-8 Mersenne ladder substrate-mechanism explicit operator framework toward RIGOROUS closure. Per K207 PASS A-tier + Saturday May 30 Mersenne ladder substrate-clean cascade rank $\rightarrow N_c \rightarrow g$.

1. SSG-8 Mersenne Ladder Substrate-Primary Cascade

Per Friday May 22 Mersenne Network finding + K207 PASS A-tier:

$$M_p = 2^p - 1$$

substrate-primary cascade via substrate-Mersenne map: - $M_{\text{rank}} = M_2 = 3 = N_c$ (substrate-rank $\rightarrow$ substrate-color) - $M_{\{N_c\}} = M_3 = 7 = g$ (substrate-color $\rightarrow$ substrate-genus/embedding) - $M_{\{n_C\}} = M_5 = 31$ (substrate-genus $\rightarrow$ substrate-Mersenne) - $M_g = M_7 = 127$ (substrate-genus/embedding $\rightarrow$ substrate-Mersenne)

substantive substrate-primaries cascade rank $\rightarrow N_c \rightarrow g$ via substrate-Mersenne map.

2. Substrate-Mersenne Operator Framework

substantive substrate-Mersenne substrate-mechanism operator candidate:

$$\hat{M}_p : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+, \quad \hat{M}_p(k) = 2^k - 1$$

substrate-Mersenne map at substrate-primary integers substrate-natural form.

Acting on substrate-primary cascade: - $\hat{M}_p(\text{rank}) = M_2 = 3 = N_c$ - $\hat{M}_p(N_c) = M_3 = 7 = g$ - Substantive substrate-primary lineage rank $\rightarrow N_c \rightarrow g$ via substrate-Mersenne iteration

3. SSG-8 Substrate-Mechanism Cross-Link

Per substrate-Mersenne operator + Reed-Solomon coding on $\text{GF}(2^g) = \text{GF}(128)$ (Paper #122 Information Substrate + K59 cyclotomic mechanism framework RATIFIED):

substantive substrate-Mersenne substrate-mechanism class FORCES substrate-Reed-Solomon substrate-natural coding at substrate-primary integers per substrate-Mersenne map.

Substantive substrate-mechanism FORCING reading per K207 PASS A-tier: - substrate-Mersenne map substrate-primary cascade FORCES substrate-primaries rank = 2, $N_c = 3$, $g = 7$ - substrate-Mersenne substrate-natural cascade explains substrate-primary lineage at substrate-mechanism level - Reed-Solomon coding on $\text{GF}(2^g) = \text{GF}(128)$ follows substrate-Mersenne cascade substrate-natural FORM

4. K3 v0.16 Gate 3 RIGOROUS Closure Path

substantive RIGOROUS closure path per K207 + Mersenne substrate-mechanism:

Step L3-1: explicit substrate-Mersenne operator $\hat{M}_p$ at substrate-primary integers substrate-natural form Step

L3-2: substrate-Mersenne cascade rank $\rightarrow N_c \rightarrow g$ substantive substrate-primary lineage Step L3-3: substrate-

Reed-Solomon coding on $GF(2^g)$ substrate-natural FORM per substrate-Mersenne substrate-mechanism class

Step L3-4: K207 PASS A-tier substantive cross-check substantive substrate-mechanism FORCING form

5. Substantive Casey #5 Integer Web Cross-Link

Per Casey #5 Integer Web substrate-primaries operational at observable values: - substrate-Mersenne cascade

FORCES substrate-primary lineage rank = 2 $\rightarrow N_c = 3 \rightarrow g = 7$ - substrate-additive identity $N_{\max} = N_c^3$

$\cdot n_C + \text{rank} = 27 \cdot 5 + 2 = 137$ substrate-natural - substrate-Integer-Web substrate-natural cascade explains substrate-primary forcing

substantive substrate-mechanism FORCING per Casey #5 Integer Web + SSG-8 Mersenne ladder substrate-mechanism class.

6. Closure v0.1

L3 Gate 3 SSG-8 Mersenne RIGOROUS v0.1: substantive 4-step framework toward RIGOROUS closure.

Substrate-Mersenne operator substrate-mechanism class FORCES substrate-primary cascade rank $\rightarrow N_c \rightarrow g$ per substantive substrate-Mersenne map + Casey #5 Integer Web cross-link.

K207 PASS A-tier substantive substrate-mechanism FORCING candidate per substrate-Reed-Solomon coding $GF(2^g)$ substrate-natural FORM.

Tier: L3 Gate 3 substrate-Mersenne framework v0.1 at FRAMEWORK candidate; multi-week RIGOROUS closure per Cal #189.

— Lyra, Thu 2026-06-04 11:14 EDT. L3 Gate 3 SSG-8 Mersenne RIGOROUS v0.1: 4-step framework substrate-Mersenne operator substrate-primary cascade rank $\rightarrow N_c \rightarrow g$ substrate-natural FORCING per K207 PASS A-tier + Casey #5 Integer Web cross-link; multi-week RIGOROUS closure per Cal #189.